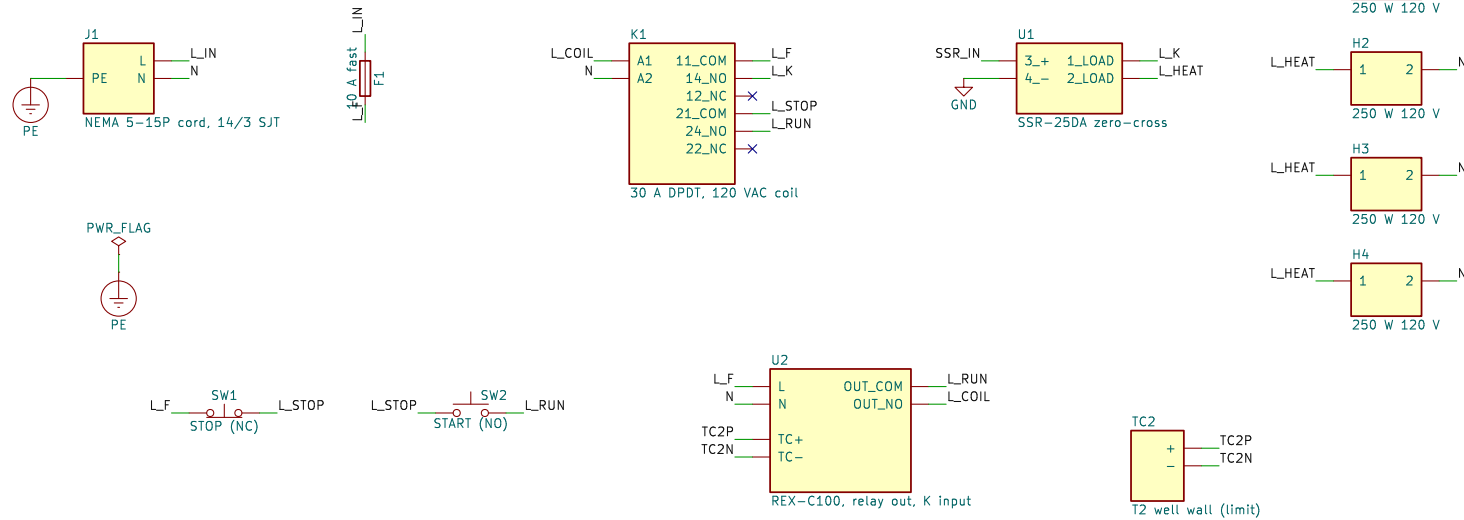


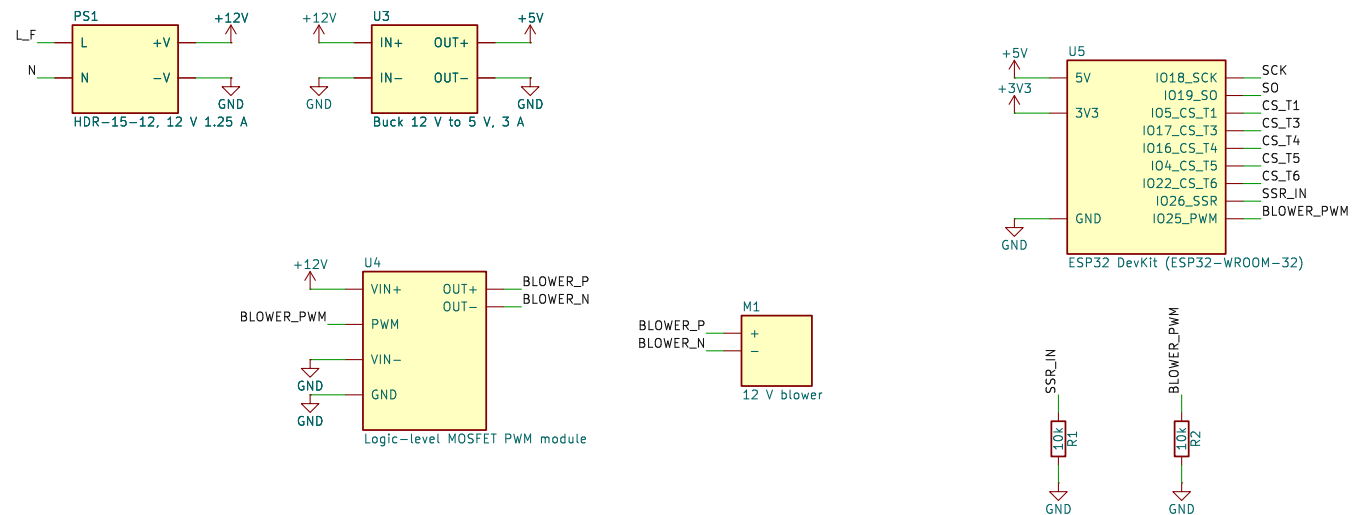
CONCEPT, NOT FOR FABRICATION

REV	DESCRIPTION	DATE	BY
P1	First issue: test article controller	2026-09-24	AC
P2	PS1 changed to a DIN-rail HDR-15-12 12 V 1.25 A supply (TBK-DWG-005)	2026-09-24	AC

120 V MAINS: heater power path and latching limit circuit



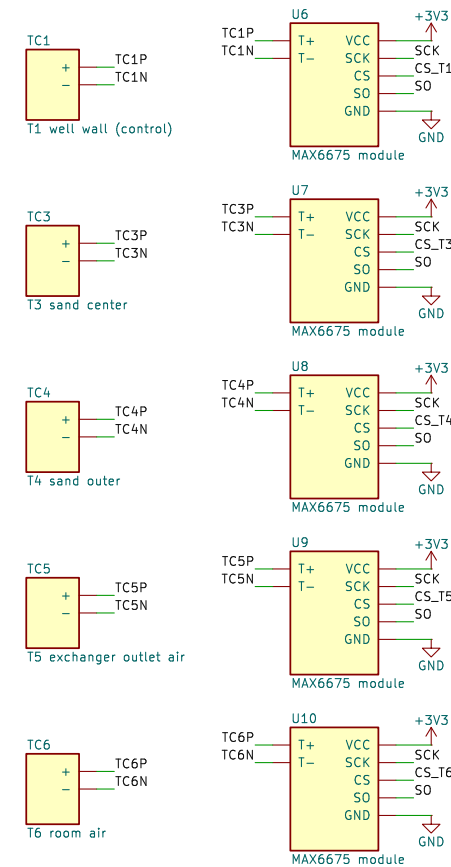
LOW VOLTAGE: 12 V, 5 V and 3.3 V. ESP32 pins per firmware/README.md



Notes

1. Cord to F1 in 14 AWG; all 120 V wiring after F1 in 16 AWG minimum, rated 105 C.
2. Heaters run only with K1 latched: START pressed, STOP closed, U2 relay closed ($T2 < 600\text{ C}$).
3. Any trip, STOP or power loss drops K1; it stays open until START is pressed.
4. U2: on-off heating action, setpoint 600 C; relay closed below setpoint, open unpowered.
5. R1 holds the SSR off through ESP32 boot and reset. If the SSR LED is dim at 3.3 V, drive U1 from +5V through an NPN (2N3904, 1k base resistor); keep R1.
6. Bond to PE: drum, wells, jacket, stack, enclosure, junction box, SSR heat sink.
7. K1, U1 and U2 terminal numbers are functional; follow each part's wiring label.
8. T2 goes only to U2, never to the ESP32, so the limit stays independent.
9. P51 and K1 mount on the DIN rail; layout in enclosure drawing TBK-DWG-005.

THERMOCOUPLES: T1, T3 to T6 on the ESP32; T2 on U2 only



THERMABRICK

Test article controller schematic

DRAWING NO. TBK-DWG-003		REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:1	UNITS n/a	PROJECTION 	DATE 2026-09-24	DRAWN Amish Chadha
MATERIAL / NOTES Electrical schematic. KiCad source in electronics/test-article; see bom/bom-test-article.csv				
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